Pointed convergence

Pointed metric space: pair (X, x)
X metric space, $x \in X$

Pointed GH-distance:

$$d_{GH}((X,x),(Y,y)) = \inf\left\{ d_H(X,Y) + d(x,y) \text{ over metrics on } X \sqcup Y \right\}$$

Pointed GH-convergence:

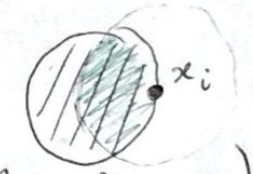
$$(X_i, x_i) \longrightarrow (X, x) \quad \text{if} \quad \forall R > 0$$

$$d_{GH}\left((\overline{B(x_i,R)}, x_i), (\overline{B(x,R)}, x) \right) \longrightarrow 0$$

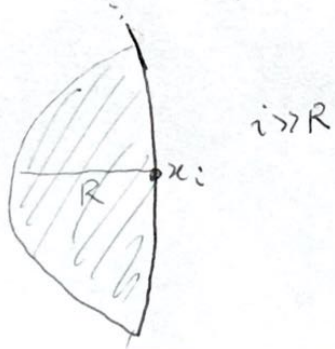
E.g. • If $(X_i, x_i) = (\underset{\mathbb{R}^2}{\overline{B(0,i)}}, 0)$

then $(X_i, x_i) \longrightarrow (\mathbb{R}^2, 0)$ as $i \to \infty$

• If $(X_i, x_i) = (\overline{B(0,i)}, (i,0))$



then $(X_i, x_i) \longrightarrow (\{x \leq 0\} \subset \mathbb{R}^2, 0)$



Convergence of maps

Suppose $X_k \to X$ in GH sense, realized
$Y_k \to Y$ by metrics on $X_k \sqcup X$
and $Y_k \sqcup Y$

and $f_k : X_k \to Y_k$

<u>Defn</u> we say f_k converges to $f : X \to Y$
if for any sequence $x_k \in X_k$ s.t. $x_k \to x \in X$
we have $\underset{\substack{\cap \\ Y_k}}{f_k(x_k)} \longrightarrow \underset{\substack{\cap \\ Y}}{f(x)}$

E.g. Let $X_k = X$, $Y_k = Y$, $f_k = f$ $\forall k$

Then $f_k \to f$ means that whenever

$x_k \to x$ in X we have $f(x_k) \to f(x)$ in Y

So f is continuous.

Defn The sequence $f_k : X_k \to Y_k$ is equicontinuous if $\forall \varepsilon > 0$ $\exists \delta > 0$ s.t. for all $x_k \in X_k$

$$f_k(B(x_k, \delta)) \subset B(f_k(x_k), \varepsilon).$$

Lemma (Arzela-Ascoli) If $X_k \to X$, $Y_k \to Y$ and $f_k : X_k \to Y_k$ are equicontinuous then $\{f_k\}$ has a convergent subsequence. Holds for pointed convergence too assuming f_k preserves basepoints.

Compactness of families of metric spaces

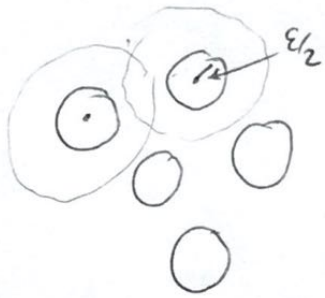
Defn If X = compact metric space, $\varepsilon > 0$

$Cap_X(\varepsilon)$ = max number of disjoint $\frac{\varepsilon}{2}$ balls in X

$Cov_X(\varepsilon)$ = min. number of ε-balls it takes to cover X

Both are decreasing functions of ε

<u>Note</u>: 1) $\mathrm{Cov}_X(\varepsilon) \le \mathrm{Cap}_X(\varepsilon) \le \mathrm{Cov}_X(\varepsilon/2)$



2) If $d_{GH}(X,Y) < \delta$ then

$$\mathrm{Cov}_X(\varepsilon + 2\delta) \le \mathrm{Cov}_Y(\varepsilon)$$

$$\mathrm{Cap}_X(\varepsilon + 2\delta) \le \mathrm{Cap}_Y(\varepsilon)$$

<u>Theorem</u> (Gromov) Let $\mathcal{C} \subset (\mathcal{M}, d_{GH})$ be a family of compact metric spaces with diameters at most D. The following are equivalent:

(1) $\mathcal{C}$ is pre-compact, i.e. every sequence in $\mathcal{C}$ has a convergent subsequence in $\mathcal{M}$.

(2) There is a function $N_1 : (0,\infty) \to (0,\infty)$ s.t. $\mathrm{Cap}_X(\varepsilon) \le N_1(\varepsilon)$ $\forall \varepsilon > 0$, $X \in \mathcal{C}$

(3) There is a function $N_2 : (0,\infty) \to (0,\infty)$ s.t. $\mathrm{Cov}_X(\varepsilon) \le N_2(\varepsilon)$ $\forall \varepsilon > 0$, $X \in \mathcal{C}$

<u>Proof</u> (1) $\Rightarrow$ (2) : Let $\varepsilon > 0$. Using (1) we can find $X_1, \dots, X_k \in \mathcal{C}$ s.t. for all $X \in \mathcal{C}$ we have $d_{GH}(X, X_i) < \varepsilon/4$ for some i

Then $$\mathrm{Cap}_X(\varepsilon) \le \mathrm{Cap}_{X_i}(\varepsilon/2) \le \underbrace{\max_i \mathrm{Cap}_{X_i}(\varepsilon/2)}_{\text{let } N_1(\varepsilon) =}$$

(2) $\Rightarrow$ (3) : Use $N_2 = N_1$

(3) $\Rightarrow$ (1) : Enough to show that $\mathcal{C}$ is totally bounded i.e. for all $\varepsilon > 0$ $\exists\, X_1, \dots, X_k \in \mathcal{M}$ s.t. for all $X \in \mathcal{C}$ we have $d_{GH}(X, X_i) < \varepsilon$ for some i.

Let $X \in \mathcal{C}$. We have $\mathrm{Cap}_X(\varepsilon/2) \le N_2(\varepsilon/2)$

$\Rightarrow X$ is within $\varepsilon/2$ of a finite subset A_X with at most $N_2(\varepsilon/2)$ elements

A_X is determined by an $N_2(\varepsilon/2) \times N_2(\varepsilon/2)$ matrix, with entries in $[0, D]$.

We can find finitely many such matrices s.t. any A_X is within $\varepsilon/2$ of one of them. We let $X_1, \dots, X_k$ be the finite metric spaces corresponding to these matrices. $\square$